Agricultural Yield Prediction System for Farmers in Kenya

AI-Driven Rainfall Forecasting & Yield Risk Classification via SMS

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AI System Design · Team 1 · March 2026

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Problem Definition

307M

Africans facing hunger daily (2025)

95%

of Kenya's farmland is entirely rain-fed

34%

drop in crop productivity since 1961

ROOT CAUSES

- Erratic rainfall → planting is pure guesswork
- Smallholders produce 80% of food with zero data access
- Climate shocks cause irreversible yield losses

OUR SOLUTION

- Predict rainfall 1–7 days; classify Low/Medium/High risk
- Agricultural advice via SMS delivery — no smartphone or internet required

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Data & Preprocessing

Climate Data

Meteor stat · 64,309 records · 30+ stations · Multi-decade

- Temp (avg/min/max) - Precipitation - Wind - Pressure
- 5 Zones: Coastal · Highlands · Nairobi · ASAL · Other

Yield Data

FAOSTAT Kenya · 800 crop-year records · 2015 – 2024

- Yield (kg/ha) · Area Harvested (ha) · 40+ crops
- Target: Yield_Category ⟶ Low / Medium / High

Preprocessing

- Zone classification & season labelling
- Rolling stats, cyclical encoding
- 40+ engineered features

Preprocessing

- Quantile binning and crop category grouping
- Temporal & Final features: Area, Years, Year Numeric, Crop Category

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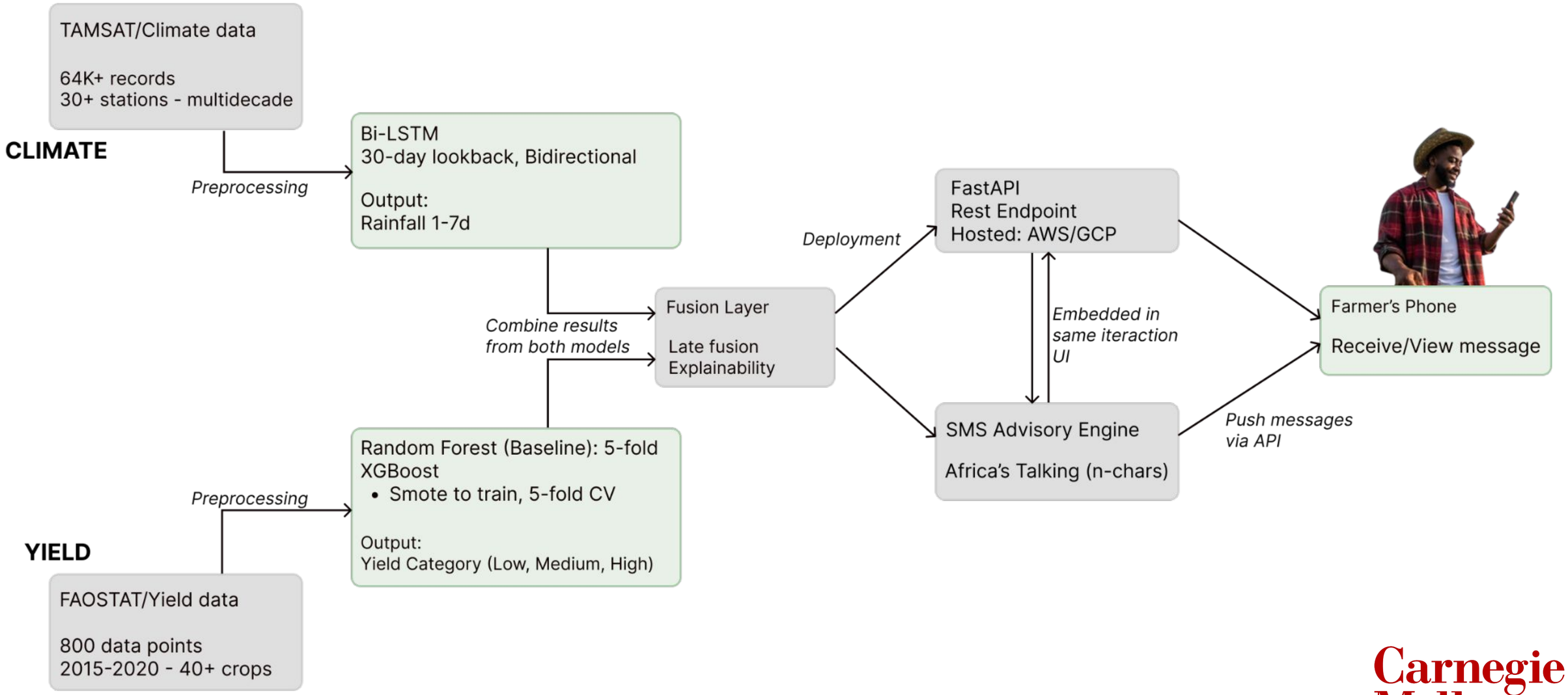
Proposed ML/DL Architecture & Deployment



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Challenges & Next Steps

Risks

- BiLSTM may struggle with irregular rainfall sequences
- Model drift — rainfall patterns shift over time
- Rare yield failures underrepresented

Ethical Concerns

- Crop bias: model trained heavily on some crops
- ASAL zones have fewer weather stations, less data
- Advice equitability across regions

PHASE 2 ROADMAP

Week 1
Model Tuning (Climate and Yield)

→

Week 2-3
FastAPI Backend, SMS Advisory Engine

→

Week 2-3
Streamlit Dashboard, Africa's Talking

→

Week 4
System Integration, Final Demo

THANK YOU

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